A Non-Intrusive Diagnostic Framework for Quantifying Ion Energy Flux Across Plasma Sheaths

Time-Resolved Optical Diagnostics Integrated with First-Principles Sheath Modeling

Team Astrophage

Nithisha V · Ajayaditya L · Denistan B · Danushika N

Problem Statement

Ion energy flux across plasma sheaths governs surface heating, erosion, and efficiency in plasma-based systems, including electric propulsion and advanced plasma reactors. While first-principles models (fluid and kinetic) estimate flux from sheath potential and plasma parameters, these predictions remain weakly validated under realistic, non-equilibrium, and time-varying conditions. Existing experimental approaches rely predominantly on intrusive probes that perturb sheath structure and distort the quantities being measured, while non-invasive diagnostics capture only partial observables insufficient to reconstruct energy deposition. Consequently, there is no widely accepted, experimentally validated framework for quantifying ion energy transfer at plasma–surface boundaries. This measurement gap introduces persistent uncertainty in predicting performance, lifetime, and thermal loads in plasma systems, limiting development of reliable diagnostic tools and constraining optimization of plasma-driven energy technologies.

Solution

This work develops a non-intrusive diagnostic framework to infer ion energy flux across plasma sheaths by combining time-resolved optical measurements with first-principles modeling. The approach leverages complementary diagnostics—optical emission spectroscopy, laser-induced fluorescence, Thomson scattering, and interferometry—to obtain partial observables such as electron temperature, density, and ion velocity distributions without perturbing the sheath. These measurements are integrated with sheath theory and kinetic simulations to construct a constrained inverse framework that reconstructs energy flux components at the plasma–surface boundary. The effort will begin with validated analytical and simulation models, followed by controlled laboratory experiments, and scale to high-resolution measurements using advanced laser and vacuum infrastructure. The objective is an experimentally grounded pathway for quantifying sheath energy transfer under realistic plasma conditions.

Innovation and Technical Approach

The central innovation replaces intrusive probes with a physics-constrained inverse framework that fuses multiple non-perturbing optical diagnostics to reconstruct ion energy flux—a quantity not directly measurable. Rather than relying on a single observable, it integrates electron temperature, density, and ion velocity distributions from complementary techniques (OES, LIF, Thomson scattering, interferometry), regularizing the inversion with sheath theory and kinetic simulations. Time-resolved acquisition captures non-equilibrium and transient dynamics inaccessible to conventional methods. This fusion of multi-diagnostic data with first-principles modeling transforms partial, indirect measurements into validated, spatially and temporally resolved energy-deposition estimates at the plasma–surface boundary.

Basic Commercialization or Implementation Potential

The framework offers direct value to electric propulsion developers and plasma-reactor manufacturers seeking accurate lifetime, erosion, and thermal-load predictions without disturbing operation. As a non-intrusive diagnostic package, it can be commercialized for thruster qualification, semiconductor process control, and fusion-relevant studies. By replacing perturbing probes with validated optical inference, it enables reliable in-situ monitoring and design optimization across plasma-driven energy technologies.